

Problem Statement & Team Details

SMART INDIA HACKATHON 2026

Problem Statement: SIH26101

Problem Statement ID – SIH26101

Problem Statement Title – Competency Gaps in India's Official Statistical System (MoSPI / NSSO)

Theme – Smart Education

PS Category – Software

Team ID – 0000

Team Name – Null Pointers

EduNexus AI Platform

AI-enabled personalized capacity-building platform that bridges competency gaps in India's Official Statistical System through deterministic mastery tracking, iGOT Karmayogi resource mapping, and automated PDF-to-quiz synthesis.



CORE ARCHITECTURAL PRINCIPLE: Right Person → Right Training → Right Time → Right Reason



❖ WHY THIS PROBLEM?

THE PROBLEM

Thousands of Statistical Officers, Survey Investigators, and Analysts across MoSPI, NSSO, and DES collect the macroeconomic, census, and sample-survey data that directly shapes national policy and budget decisions — high-stakes work with zero room for error.

CRITICAL BOTTLENECK

Data inaccuracies caused by outdated field methods lead to faulty national forecasts and delayed policy interventions — the cost of a poorly-trained cadre is paid at the national level.

ONE-SIZE-FITS-ALL FAILURE

Traditional training models enroll all officers in the same uniform basic program, regardless of individual strengths and weaknesses — wasting valuable working hours on skills already mastered.

Real-world example

Officer Ramesh: Data Visualization: 80% (mastered)

- **Sampling Techniques: 55% (severe gap).**

A generic course wastes his time on visualization while the real gap goes unaddressed.



❖ Existing Solutions

01

Manual Gap Auditing Does Not Scale

Across 10,000+ distributed cadre officers, it is humanly impossible for HR/Trainers to manually assess individual topic weaknesses across micro-competencies.

The real gap: No deterministic, explainable mathematical score exists to rank and prioritize officer training needs.

02

iGOT Has Content, Missing Personalization

iGOT Karmayogi hosts thousands of high-quality modules, but officers suffer from “search fatigue” and lack guidance on exactly what matches their skill gap.

The real gap: Officers are forced to self-diagnose without a continuous, competency-driven recommendation map.

03

Manual Quiz Creation Is Bottlenecked

When MoSPI releases new survey manuals (e.g. PLFS, ASI guidelines), creating practice questions and MCQ assessments takes instructors weeks of manual work.

The real gap: No automated PDF-to-validated-MCQ ingestion pipeline exists to create instant, contextual evaluations.

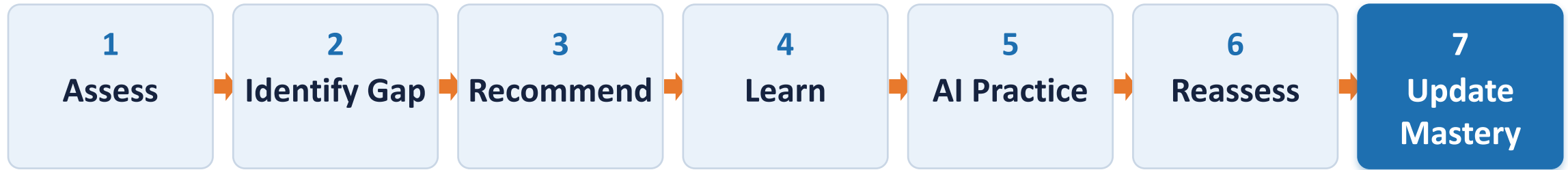
EduNexus AI acts as an intelligent personalization layer on top of iGOT Karmayogi — not a replacement for it.



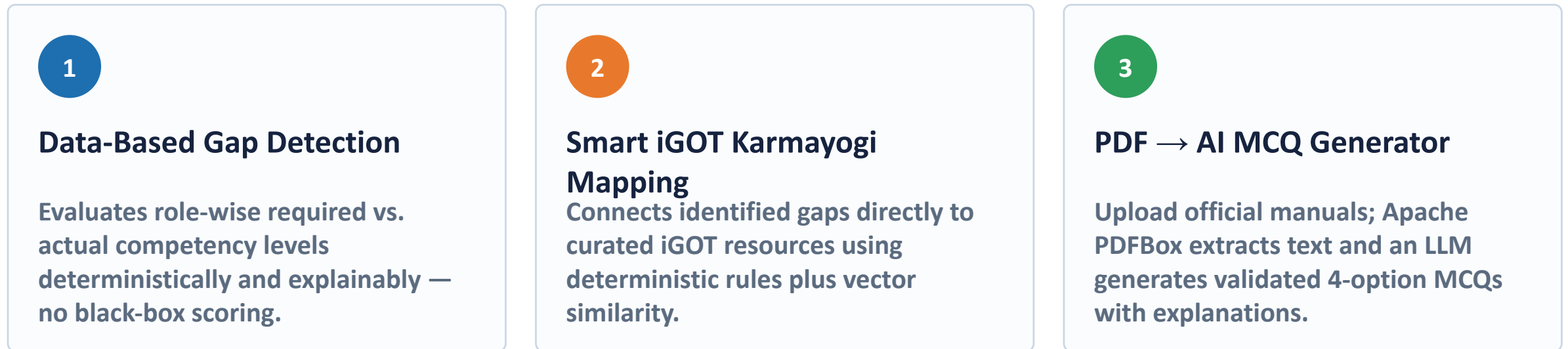
A closed-loop autonomous system delivering targeted capacity building



❖ Proposed Solution



CONTINUOUS LOOP — mastery feeds back into the next assessment cycle



Deterministic Mastery Calculation: $\text{Mastery} = (\text{Correct Answers} \div \text{Total Attempted}) \times 100\%$ | $\text{Gap} = \text{Required} - \text{Current}$



Engineered for reliability, security, and enterprise government scalability



❖ TECHNICAL ARCHITECTURE & STACK

Backend & Security

- **Node.js + Express.js + TypeScript**

Scalable and efficient backend development with REST APIs and modular architecture.

- **JWT Authentication + bcrypt**

Secure authentication, password hashing, and role-based authorization.

- **Prisma ORM + Supabase PostgreSQL**

Type-safe database access, relational modeling, and reliable data storage.

Data & Intelligence

- **Supabase PostgreSQL**

Relational storage for assessments, competency scores, mastery tracking, and learning resources.

- **PDF Parser + OpenRouter AI**

Automated document processing, content extraction, and AI-powered MCQ generation.

- **Recommendation Engine & iGOT Adapter**

Competency gap detection with personalized resource recommendations and seamless iGOT integration.

Frontend & DevOps

- **React.js + Tailwind CSS**

Responsive and user-friendly interface for analytics, assessments, and learning workflows.

- **Docker**

Containerized deployment for consistent development and demo environments.

- **Git & Git Hub**

Version control, collaboration, and streamlined project management.

Fallback reliability: a local fallback question bank and cached resources ensure 100% demo uptime.



❖ SCALABILITY, SUSTAINABILITY & IMPACT

100%

Personalized Training
Pathways

85%

Reduction in Quiz Creation
Time

10k+

Officers Scalable via Cloud
Architecture

5%

AI Hallucination Risk

Deployment & Sustainability Plan

Phase 1 — NSSTA & MoSPI HQ Pilot

Initial deployment with 200 statistical officers, validating topic models for survey methods and sampling.

Phase 2 — Pan-India State Statistical Cadres

State DES integration with role-specific competency benchmarks and regional localized modules.

Enterprise Governance & Audit

Full audit logs for compliance, data privacy, and verifiable continuous professional development (CPD).



◆ Future Scope & Limitations

<u>FUTURE SCOPE & ENHANCEMENTS</u>	<u>CURRENT LIMITATIONS</u>
• Live production iGOT Karmayogi API single-sign-on integration. • Adaptive difficulty testing based on Item Response Theory (IRT). • Multi-lingual support (Hindi & regional languages) for field investigators. • Offline-first PWA mode for remote survey officers in rural areas.	• iGOT karmyogi live API not yet integrated; currently using mock adapter for demo. • LLM generated MCQ's require human validation. • PostgreSQL +pgvector may face performance degradation beyond 1Lakh+ current user. • Platform currently supports English only; no regional language support for field investigators.